

#HW 7

● Graded

Student

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Total Points

92 / 100 pts

Question 1

(no title)

10 / 10 pts

✓ - 0 pts Correct

- 2 pts del T is same in both rankine and fahrenheit scales.
- 2 pts 6 inches is 0.5 ft
- 5 pts $Q = K A \Delta T / L$.
- 3 pts check the calculation properly.
- 1 pt total heat loss in 10 hours not calculated
- 10 pts not answered

Question 2

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts $Q_{\text{total}} = Q_{\text{convection}} + Q_{\text{radiation}}$; step missed
- 2 pts check the calculation properly
- 15 pts question hasn't been answered

Question 3

(no title)

12 / 15 pts

- 0 pts Correct

✓ - 3 pts $W = m (g/gc)$

- 3 pts $m = PV/RT$
- 2 pts check the units/ calculation properly
- 10 pts didnt answer the question properly

Question 4

(no title)

15 / 15 pts

✓ - 0 pts Correct

- 2 pts calculation error

- 5 pts $Q = m C_p \Delta T$

- 5 pts incomplete calculation

Question 5

(no title)

15 / 15 pts

✓ - 0 pts Correct ; 634 K or 361 C

- 3 pts calculation error

Question 6

(no title)

12 / 15 pts

- 0 pts Correct

- 5 pts incomplete solution

✓ - 3 pts Calculation mistake/ error.

Question 7

(no title)

13 / 15 pts

- 0 pts Correct

- 3 pts calculation error in work done

- 2 pts calculation error in Q

✓ - 2 pts part b calculation error/ not completed.

Homework 07

1. Problem 2.13.35

$$\dot{Q} = k \cdot A \left[\frac{T_2 - T_1}{L} \right]$$

$$\frac{6 \text{ in}}{12} = 0.5 \text{ ft}$$

$$A = 20 \cdot 10 = 200 \text{ ft}^2$$

$$68^\circ\text{F} - 40^\circ\text{F} = 28^\circ\text{R}$$

$$k = 0.04 \text{ Btu/h} \cdot \text{ft} \cdot ^\circ\text{R}$$

$$\dot{Q} = \frac{0.04 (200) (28)}{0.5} = \frac{448 \text{ Btu}}{\text{h}} \cdot 10 \text{ h} = \boxed{4480 \text{ Btu}}$$

2. Problem 2.13.41

$$\dot{Q} = \text{convection} + \text{radiation}$$

$$\dot{Q}_{\text{conv}} = h \cdot (T_b - T_a)$$

$$h = 10 \text{ W/m}^2 \cdot \text{K}$$

$$T_b = 47^\circ\text{C} + 273.15 = 320.15$$

$$T_a = 27^\circ\text{C} + 273.15 = 300.15$$

$$\dot{Q}_{\text{conv}} = 10 \cdot (320.15 - 300.15) = 200 \text{ W/m}^2$$

$$\dot{Q}_{\text{rad}} = \varepsilon \cdot \sigma (T_b^4 - T_a^4)$$

$$= 0.93 \cdot 5.67 \times 10^{-8} \cdot (320.15^4 - 300.15^4) = 126 \text{ W/m}^2$$

$$\dot{Q} = 200 + 126 = \frac{326 \text{ W/m}^2}{1000} = \boxed{0.326 \text{ kW/m}^2}$$

3. Problem 3.21.71

$$PV = nRT$$

$$V = 110 \times 60 \times 25 = 165,000 \text{ ft}^3, 28.317 = 4,672,305 \text{ L}$$

$$P = 1 \text{ atm}$$

$$T = (70^\circ\text{F} - 32) \cdot \frac{5}{9} = 21.11 + 273.15 = 294.26 \text{ K}$$

$$R = 0.08205 \text{ L} \cdot \text{atm} / \text{mol} \cdot \text{K}$$

$$(1) (4,672,305) = n (0.08205) (294.26)$$

$$n = \frac{193,518 \text{ mol}}{1 \text{ mol}} \cdot \frac{28.97 \text{ g}}{453.6 \text{ g}} = 12,360 \text{ lb}$$

$$W = m \cdot g = 12,360 \cdot 32 = 395,500 \text{ lb} \cdot \text{ft}$$

4. Problem 3.21.79

$$PV = mRT$$

$$1 \text{ bar} = 100 \text{ kPa}$$

$$V = 2 \text{ m}^3$$

$$m = 2 \text{ kg}$$

$$R = 8.314 \text{ kJ} / \text{kmol} \cdot \text{K} / 32 \text{ kg} / \text{kmol} = 0.25981 \text{ kJ} / \text{kg} \cdot \text{K}$$

$$2(100) = 2(0.25981) T_1$$

$$T_1 = 384.90 \text{ K}$$

* V doubles, T doubles; $2 \cdot T_1 = 769.8 \text{ K}$

$$c_p = \frac{k \cdot R}{k-1} = \frac{1.35(0.25981)}{1.35-1} = 1.002$$

$$Q = m \cdot c_p (\Delta T)$$

$$Q = 2 \cdot 1.002 (384.91) = 771.44 \text{ kJ}$$

5. Problem 3.21.84

$$T_1 = 430 + 273.15 = 703.15 \text{ K}$$

$$Q = m \cdot c_v (\Delta T)$$

$$Q = 425 \text{ W} \cdot (10 \text{ min} \cdot 60 \text{ s}) = 255,000 \text{ J} \rightarrow 255 \text{ kJ}$$

$$m = 5 \text{ kg}$$

$$c_v = \frac{R}{\gamma - 1} = \frac{0.25981}{1.35 - 1} = 0.7423$$

$$-255 = 5 (0.7423) (T_2 - 703.15)$$

$$-255 = 3.7115 T_2 - 2,609.74$$

$$T_2 = 634.53 \text{ K} - 273.15 = \boxed{361.38^\circ \text{C}}$$

6. Problem 3.21.90

$$V = 0.29 \text{ ft}^3$$

$$T_1 = 560^\circ \text{R}$$

$$P = 18.16 \text{ lbf/in}^2 \cdot (12^2) = 2592 \text{ lbf/ft}^2$$

$$W = 1.7 \text{ Btu} \cdot 778.17 = 1,322.889 \text{ ft} \cdot \text{lbf}$$

$$R = 53.35 \text{ ft} \cdot \text{lbf/lbm} \cdot \text{R}$$

$$PV = mRT$$

$$(2592)(0.29) = m (53.35)(560)$$

$$m = 0.02516 \text{ lbm}$$

$$c_v = \frac{R}{\gamma - 1} = \frac{53.35}{1.4 - 1} = 133.375$$

$$W = m \cdot c_v (\Delta T)$$

$$1322.889 = (0.02516 \cdot 133.375) (T_2 - 560^\circ \text{R})$$

$$1322.889 = 3.355 T_2 - 1479.12$$

$$\boxed{T_2 = 954^\circ \text{R}}$$

7 Problem 3.21.91

a)

$$PV = mRT$$

$$0.7 \text{ bar} = 70 \text{ kPa}$$

$$0.262 \text{ m}^3$$

$$R = 0.1889$$

$$T = 280 \text{ K}$$

$$70(0.262) = m(0.1889)(280)$$

$$m = 0.3467 \text{ kg}$$

$$P_1 V_1^n = P_2 V_2^n$$

$$V_2 = V_1 \left(\frac{P_1}{P_2} \right)^{\frac{1}{n}}$$

$$V = \frac{mRT}{P}$$

$$\frac{mRT_2}{P_2} = \frac{mRT_1}{P_1} \left(\frac{P_1}{P_2} \right)^{\frac{1}{n}}$$

$$\frac{T_2}{P_2} = \frac{T_1}{P_1} \left(\frac{P_1}{P_2} \right)^{\frac{1}{n}}$$

$$T_2 = \frac{T_1 P_2}{P_1} \left(\frac{P_1}{P_2} \right)^{\frac{1}{n}} = T_1 \left(\frac{P_2}{P_1} \right)^{\frac{n-1}{n}}$$

$$T_2 = 280 \left(\frac{1100}{70} \right)^{\frac{1.25-1}{1.25}} = 485.73 \text{ K}$$

$$W = \frac{mR(T_2 - T_1)}{1 - n} = \frac{0.3467 \cdot 0.1889 \cdot (485.73 - 280)}{-0.25} = -54 \text{ kJ}$$

$$Q = \Delta U + W = m c_v (\Delta T) + W$$

$$= (0.3467)(0.657)(485.73 - 280) + (-54)$$

$$Q = -7.14 \text{ kJ}$$

$$b) \quad 280 \text{ K} = 6,369 \text{ kJ/kmol} \cdot \left(\frac{0.3467}{u_1} \right) = 50.18$$

$$485.73 \text{ K} = 12,979 \text{ kJ/kmol} \cdot \left(\frac{0.3467}{u_2} \right) = 102.27$$

$$\Delta U = m(u_2 - u_1) = 0.3467(102.27 - 50.18) = 18.06 \text{ kJ}$$

$$W = -54 \text{ kJ}$$

$$Q = \Delta U + W = -54 + 18.06 = -35.94 \text{ kJ}$$